Project Lawi

Investment Opportunity

April 2026

Investment Highlight

Integrated platform with delivering immediate cash flow with scalable, bankable infrastructure growth

Strategic Circular Ecosystem & Cost Leadership

- Fully integrated value chain (RDF > Power > Activated Carbon).
- Drives superior margins and establishes a formidable barrier to entry through unbeatable cost advantages.

Immediate Cash Flow & Efficient Capital Deployment

- Day-1 cash flow from acquisition of operating UPT assets with zero construction risk
- Strong operating cash flow supports project financing and accelerates portfolio scale-up.

Secured Revenue & High Bankability Profile

- Long-term off-take agreements with top-tier government and corporate counterparties (e.g., PEA, GULF).
- Predictable and stable cash flows are high project bankability

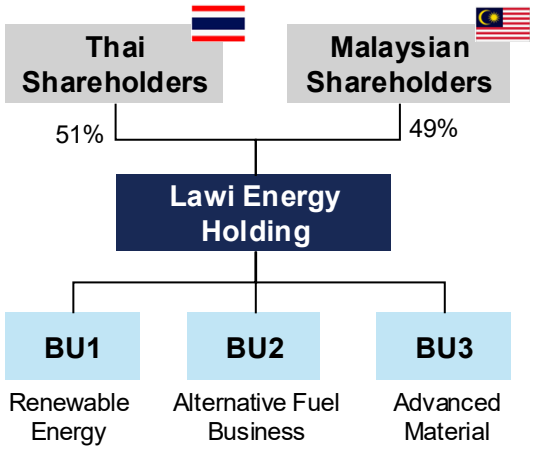
Operational Excellence & Enhanced Profit Margins

- In-house O&M capabilities driving performance optimization and additional fee income.
- Directly boosts net profits and shareholder return.

ESG-Driven Growth & Access to Low-Cost Green Capital

- Aligned with decarbonization trends and national energy policy (PDP Plan).
- Access to green financing instruments (Green Bonds, Sustainability-Linked Loans)

Project Structure



Capital Structure

Project Cost	4,900 MB	100%
Equity	2,450 MB	[50] %
Debt	2,450 MB	[50] %

Preliminary Investment Return

PIRR (%)	10.5 – 12.0 %
EIRR (%)	14.5 – 16.0%
Payback	7-10 Years

Key Project Management Team

An integrated leadership team combining EPC execution, project development, technical expertise, and financial structuring to deliver scalable waste-to-energy solutions

EPC & Engineering Lead

Rathakrishnan Arumugam

Experience:

- Project Director, Lawi Engineering (Malaysia)
- **20+ years** in renewable energy engineering and EPC execution
- Expertise in thermal waste systems and biomass utilization

Project Development Lead

Suntad Yingyong

Experience:

- Currently, he is Managing Director at Global Green Holding Co., Ltd. and Green Parawood Co., Ltd.
- Over 45 years of professional experience in Civil Engineering, factory management, and industrial construction.
- Extensive expertise in managing large-scale operations across the food processing (canned tuna), renewable energy, and construction sectors.

Technical & Policy Lead

Dr. Udomsak Kaewsiri

Experience:

- Currently, he is Founder & Executive Director at Lawi Engineering (Thailand) and KK Green Co., Ltd.
- Over 30 years of experience in biomass power generation.
- Expertise in engineering design, CO_2 mitigation, and the governance of large-scale renewable energy infrastructure.
- Ph.D. candidate in Computer Engineering and Financial Technology, integrating industrial practice with advanced academic research

Finance & Strategy Lead

Manaphat Asakit

Experience:

- Strategic Advisor across infrastructure and energy projects
- **20+ years** in banking, corporate strategy, and project finance
- Experience with leading financial institutions and EPC companies

Industry Opportunity in Thailand

Thailand presents a strong structural opportunity across waste management, energy demand, and industrial decarbonization

Waste Management Gap

Over **27 million tons** of Waste annually

- Low landfill efficiency and limited waste processing capacity
- Increasing pressure to reduce landfill usage

Growing Energy Demand

growth **4-5% p.a.** in Energy Generation from Renewable Energy

- Rising electricity demand driven by industrial growth
- Required for stable base-load power
- Waste-to-energy provides reliable and continuous generation

Decarbonization & Policy Support

51% Renewable Energy Targets within 2037

- Government targets for RE expansion from 36% to 51% of total power generation (PDP plan 2024)
- Supportive policies for WTE and biomass projects
- Net-zero commitments driving energy transition

Industrial Demand & Import Substitution

Over **18 million tons** of Coal imported annually

- RDF provides cost-competitive alternative to coal
- Reduces carbon intensity for industrial users
- Biofuels emerging is a key substitute for Coal imported

WTE Integrated Value Chain

Vertical Integration from waste sourcing to activated carbon captures margin at every stage of the value chain

Waste Collection

Biomass, MSW, Wood sourcing

RDF Processing

Shredding, Drying and Pelletizing

WTE Power Generation

98% combustion efficiency

Activated Carbon

Emission Control

Multiple Revenue Streams

Power · Fuel · Carbon

Margin Protection

- Internalise fuel supply chain
- Protect WTE operating margins
- Eliminate third-party feedstock risk

Commercial Advantage

- Low-cost alternative fuel for industrials
- Monetise waste across multiple streams
- Capture feedstock-to-product value

Scalability

- Modular design for rapid deployment
- ESG-compliant circular model
- Replicable across Southeast Asia

Investment & Growth Strategy

Prioritizing M&A to generate immediate cash flow, enabling reinvestment into high-growth greenfield opportunities

Strategic M&A

1. **Accelerate scale portfolio** through acquisition of operating assets
2. **Drive immediate cash flow** via integration of proven operating assets.

Greenfield Development

1. Establish a scalable, long-term growth platform through disciplined greenfield investments
2. **Targeted Growth Projects**: Allocate capital into targeted, high-growth projects (e.g. RDF Plants, Activated carbon, PPA)
3. **Solar PPA (Future)**: Develop and operate Solar portfolio (Private & Government), integrated with BESS and microgrid solutions

"Acquired → Generate Cash → Reinvest → Scale up"

Current Investment Pipeline

Executing M&A to generate immediate cash flow, funding a scalable greenfield growth pipeline

Strategic M&A → **Greenfield Development**

BU 1: Renewable Energy

UPT Biomass Power Plant Acquisition

Acquisition of WTE Power Plant project (PPA ended May 2039)

Current Status	Operating
Install Capacity	9.9 MW
Contract Capacity	8 MW
Remaining Contract	13 Years PPA
Revenue	Electricity Selling
COD	May 2019
Acquisition Cost	[800] MB
Equity Injected	[400] MB

Waste-to-Energy Development

Development of WTE power plant projects 24 MW at Nonthaburi (Green field Project)

Current Status	Plan
Install Capacity	30 MW
Contract Capacity	24 MW
Contract	[20] Years PPA
Revenue	Electricity Selling
COD	TBC
Project Cost	[3,500] MB
Equity Injected	[1,750] MB

BU 2: Alternative Fuel

RDF Plants

RDF machinery installation to produce alternative fuel for power generation to WTE Power Plant and Industrial Users

Current Status	Plan
Full Capacity	250K Ton/Day
Revenue	RDF Selling
COD	End of 2026
Customer Focus	WTE Plant / Coal replacement
Project Cost	[500] MB
Equity Injected	[250] MB

BU 3: Advanced Material

Activated Carbon (AC)

Production of activated carbon from coconut/wood biomass to reduce chemical consumption in WTE Plants

Current Status	Plan
Full Capacity	200 Kg/hr
Revenue	AC Selling
COD	End of 2026
Customer Focus	WTE Plant / Consumer Products
Project Cost	[100] MB
Equity Injected	[50] MB

Future Project Investment Pipeline

Powering green energy to customer - decarbonization

Solar + BESS (Optional) Project

ESCO Model (G2G)
Contract will be engaged between customer – PEA/MEA - investor

Solar Rooftop / Floating

PEA / MEA

Government Organization i.e. Hospital, University

Private PPA (P-PPA) – (B2B)
Contract will be engaged directly between customer –investor

Solar Rooftop / Floating

BESS (Optional)

Private Company

PPA (B2G) – Policy PPA
Contract will be engaged directly between customer – EGAT/PEA/MEA

Solar Farm

BESS (Optional)

EGAT/ PEA / MEA

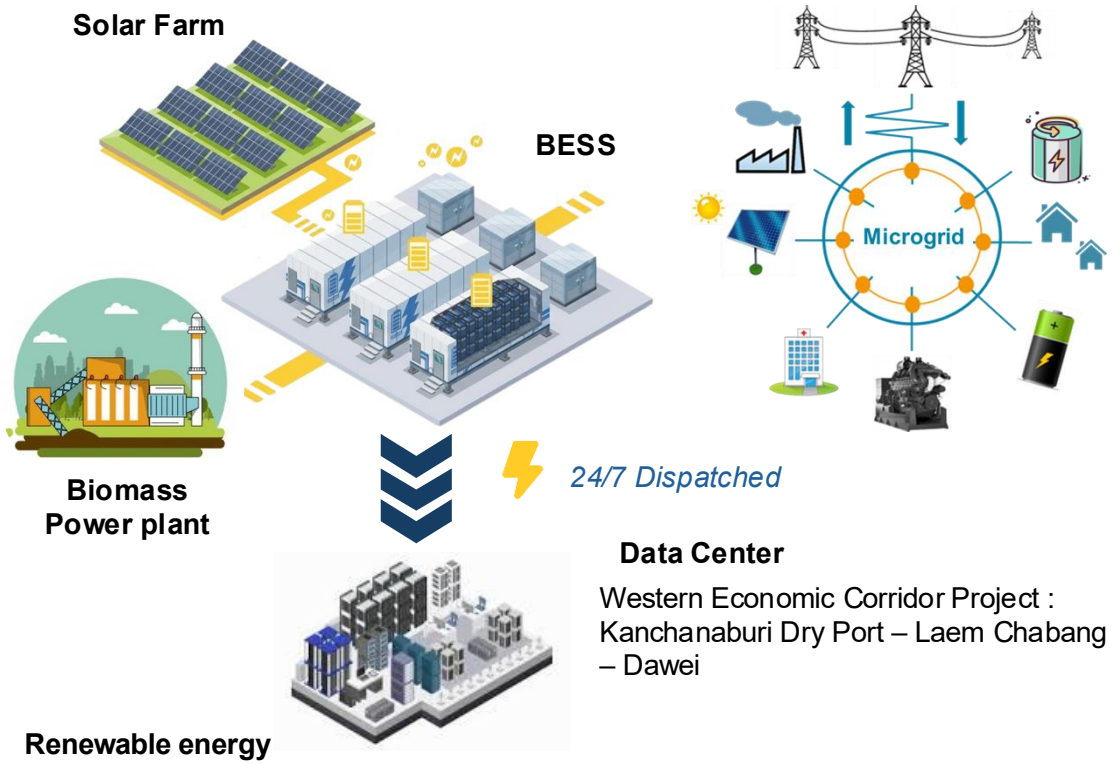
Project Information

Project Overview	
Customer Focus	Private Company / Public Government / PEA-MEA-EGAT
Capacity Target	10-20 MW
Pricing Mechanism	Discount from PEA/MEA / FiT
Contract	15-25 Years / 20 Years
Prelim Return/Investment	
Project Cost	c. [250-500] MB
Equity Injection	c. [125-250] MB
PIRR%	G-PPA [8-10%] P-PPA [10-12%]
EIRR%	G-PPA [10-12%] P-PPA [12-14%]
Financial Highlight (Year)	
Revenue	
Annual Generation	[13-26] GWh
Electricity Price	2.8-3.2 Baht/kWh
Revenue	G-PPA [25-50] MB P-PPA [30-60] MB
EBITDA	G-PPA [20-40] MB P-PPA [24-48] MB

Future Project Investment Pipeline

Powering data centers with contracted 24/7 clean energy (Direct Supply to Data Center)

Biomass + Solar Farm + BESS Project



Western Economic Corridor Project : Kanchanaburi Dry Port – Laem Chabang – Dawei

Renewable energy

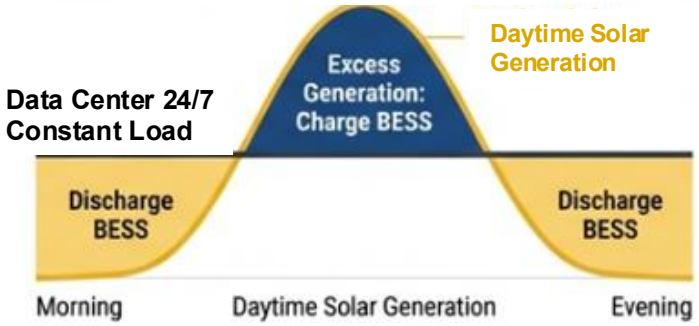
- Biomass Power Plant 2x10 MW
- Solar Farm 50-60 MW
- Battery Energy Storage System 50-60 MW

Utilities Infrastructure

- 22 kV Electrical Substation
- 22 kV Distribution Line
- Electrical Network

Why Biomass + Solar farm + BESS Wins

The 24/7 Firming Curve



	Biomass	Standard Solar	BESS	Biomass + Solar + BESS
Generation Profile	Base Load (All day)	Daytime only	Depend on Generator	**Firm, Baseload-like (24/7 Dispatchable)**
Off-taker Alignment		Misaligned with 24/7 demand	Fulfill load from solar uncertainty	**Fully aligned with 24/7 demand**
PPA Pricing	Medium	Low	High	**Below Grid parity**

"Transforming intermittent solar into premium, 24/7 contracted power"

BU1 UPT Biomass Power Plant Acquisition (Operating)

Acquiring an operating biomass power plant with immediate cash flow and contracted revenue

Project Highlight

Operating Asset with Proven Track Record

Plant has been operating since 2019 with stable performance

Long-Term Contracted Revenue (PPA)

Electricity sold under long-term PPA with remaining 14 years contract life

Immediate Cash Flow Recognition

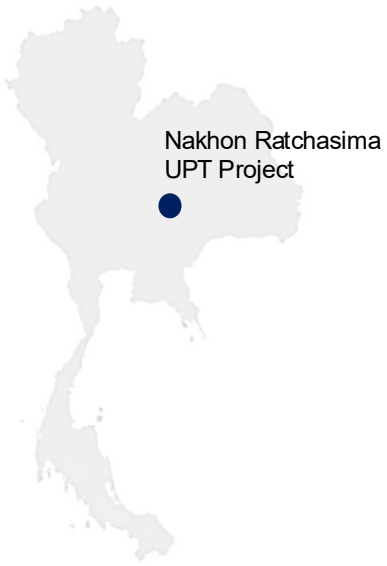
Generates revenue from day one post-acquisition

No Development Risk

No construction risk because the project already operated and recognized performance

Location

Located in Korat with access to consistent wood biomass supply to ensure stable fuel sourcing.



Processing

Converts wood biomass into stable electricity through proven thermal generation

Wood Biomass

↓

Thermal Conversion

↓

Electricity Generation

↓

Emission Control & Residue Treatment

Project Information

Project Overview	
Project Location	Nakhon Ratchasima
Current Status	Operating since May 2019
Install Capacity	9.9 MW (Contracted 8 MW)
Raw Material	Woodchip Biomass
Pricing Mechanism	Feed in Tariff + Adj.
Contract	Long-Term PPA
COD	May 2019
Remaining Tenor	13 Years (End May 2039)
Prelim Return/Investment	
Project Cost	c. [800] MB
Equity Injection	c. [400] MB
PIRR%	[10 - 12%]
EIRR%	[13 - 17%]
Actual Financial (FE2024)	
Revenue	
Annual Generation	66 Gwh
Electricity Price	4.66 Baht/Kwh
Revenue	**308**
(-) Fuel Cost	(106)
(-) OPEX	(59)
EBITDA	**144**

BU1 WTE Development Project (Green Field)

Developing waste-to-energy plants with stable, long-term contracted revenue

Project Highlight

Base-Load Power Generation Asset

Stable and predictable electricity generation under long-term PPA

Long-Term Contracted Revenue (PPA)

20 years power purchase agreement ensures predictable income

Attractive Infrastructure Returns

Stable and recurring cash flow suitable for long-term investors

Turn Waste into Energy

Converts waste into electricity (waste management + energy generation)

Location

Located in Nonthaburi near major urban waste sources to ensure stable feedstock supply and lower transportation cost

Processing

Converts waste into stable electricity through a controlled thermal process

Waste Collection & Pre-treatment

Thermal Conversion

Electricity Generation

Emission Control & Residue Treatment

Project Information

Project Overview	
Project Location	Nonthaburi
Current Status	Plan
Install Capacity	30 MW (Contract 24 MW)
Raw Material	Municipal Solid Waste (MSW)
Pricing Mechanism	Feed in Tariff + Adj.
Contract	Long Term PPA
COD	To be confirmed (Develop)
Tenor	c. 20 Years
Prelim Return/Investment	
Project Cost	c. 3,500 MB
Equity Injection	c.1,750 MB
PIRR%	[10 - 13%]
EIRR%	[16 - 21%]
Target Performance	
Revenue	
Annual Capacity	c. 160 GWh
Electricity Price	c. 3.66 B/Kwh
Revenue	c. 637 MB
(-) Operating Cost	c. -151 MB
(-) SG&A	c. -7. MB
EBITDA	c. 480 MB

BU2 RDF Plant Projects

This project is help to secure the energy value chain by converting waste into standardized industrial fuel

Project Highlight

Integrated Waste-to-Fuel Value Chain

End-to-end process by ensuring control over quality and cost

Strong Captive Offtake Demand

Dedicated demand from WTE plants reduces market risk and enhances revenue visibility

Cost-Competitive Fuel Alternative

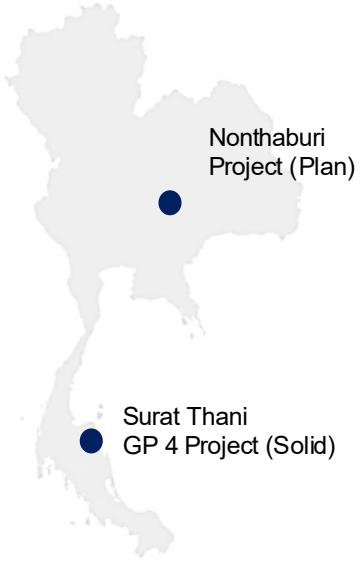
Lower and more stable cost compared to conventional fossil fuels.

Environmental & Regulatory Tailwinds

Aligned with Thailand's waste management and renewable energy policies

Location

Optimizes logistics cost and ensures reliable fuel supply to GP4 (operational) and TPCH (pipeline)



Processing

Securing Fuel Supply for WTE through Integrated Waste-to-Fuel Platform

Waste Sourcing & Management

Transportation

RDF Processing
- Crushing
- Drying
- Sieving
- Pellet Station

RDF Pellets

Project Information

Project Overview		
Project Location	Surat Thani	Nonthaburi
Current Status	Plan	Plan
End-Users	GP4 (WTE)	TPCH (WTE 24MW)
Raw Material	Waste	Waste
Contract	Long-Term	Long-Term
COD	End of 2026	End of 2027
Prelim Return/Investment		
Project Cost	c. 500 MB	
Equity Injection	c. 250 MB	
Expected PIRR%	[10 – 14%]	
Expected EIRR%	[12– 18%]	
Annual Target Financial[1/]		
Annual Output	c. 250 K Ton	
Selling Price	c. 80KB per ton	
Revenue	**c. 300 MB**	
(-) Cost	c. -223 MB	
(-) OPEX	c. -10 MB	
EBITDA	**c. 64 MB (21.3%)**	

Remark: 1/ The number is preliminary analysis which is subject to further due diligence process

BU3 Activated Carbon Project

Strategic Margin Expansion is transforming low-value biomass into essential industrial high value input

Project Highlight

Circular Economy Value Creation

Transform low-value biomass into high-margin activated carbon

Strategic Location for Material Sourcing

Location sourcing ensures stable, low-cost, and diversified raw material supply

Revenue Stability

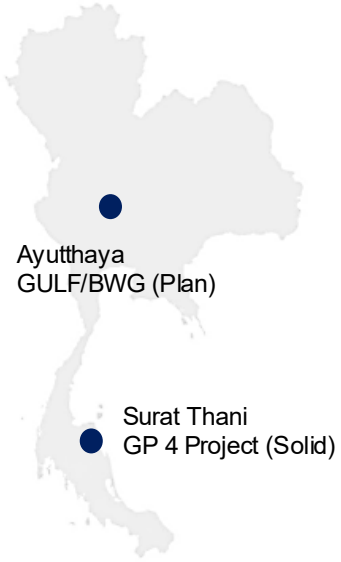
Long-term contracts with WTE Plant underpin demand visibility and revenue stability

Scalable Platform for Expansion

Modular capacity (100 kg/hr per site) with potential to scale across Thailand and ASEAN

Location

Strategic position in Ayutthaya and Surat Thani are optimizing feedstock proximity



Processing

Transforming low-cost biomass into high-margin industrial-grade activated carbon

Low-Cost Input

Raw Coconut / Wood Biomass

Thermal Processing

High Value Output

Chemical-Grade Activated Carbon

Project Information

Project Overview		
Project Location	Surat Thani	Ayutthaya
Current Status	Development	Plan
End-Users	GP4	GULF, BWG
Raw Material	Wood / PKS	Coconut Biomass
Contract	Long-Term	Long-Term
COD	End of 2026	End of 2027
Prelim Return/Investment		
Project Cost	c. 100 MB	
Equity Injection	c. 50 MB	
Expected PIRR%	[12 – 20%]	
Expected EIRR%	[15 – 25%]	
Annual Target Financial[1/]		
Annual Output	c. 1,000 Ton (80% Utilization)	
Selling Price	c. 70KB per ton	
Revenue	**c. 70 MB**	
(-) Raw Material	c. -20 MB	
(-) OPEX	c. -11 MB	
EBITDA	**c. 25 MB (28.5%)**	

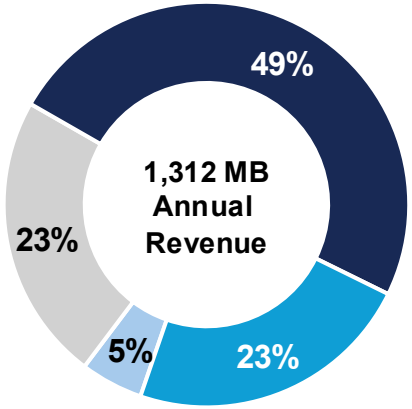
Remark: 1/ The number is preliminary analysis which is subject to further due diligence process

Revenue Mix and EBITDA Contribution

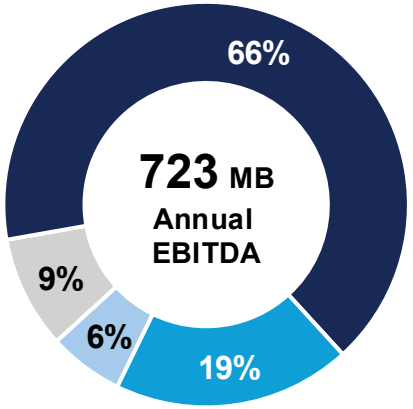
Balanced portfolio with EBITDA driven by high-margin and contracted assets

Total Target Revenue and EBITDA generation

Target Revenue



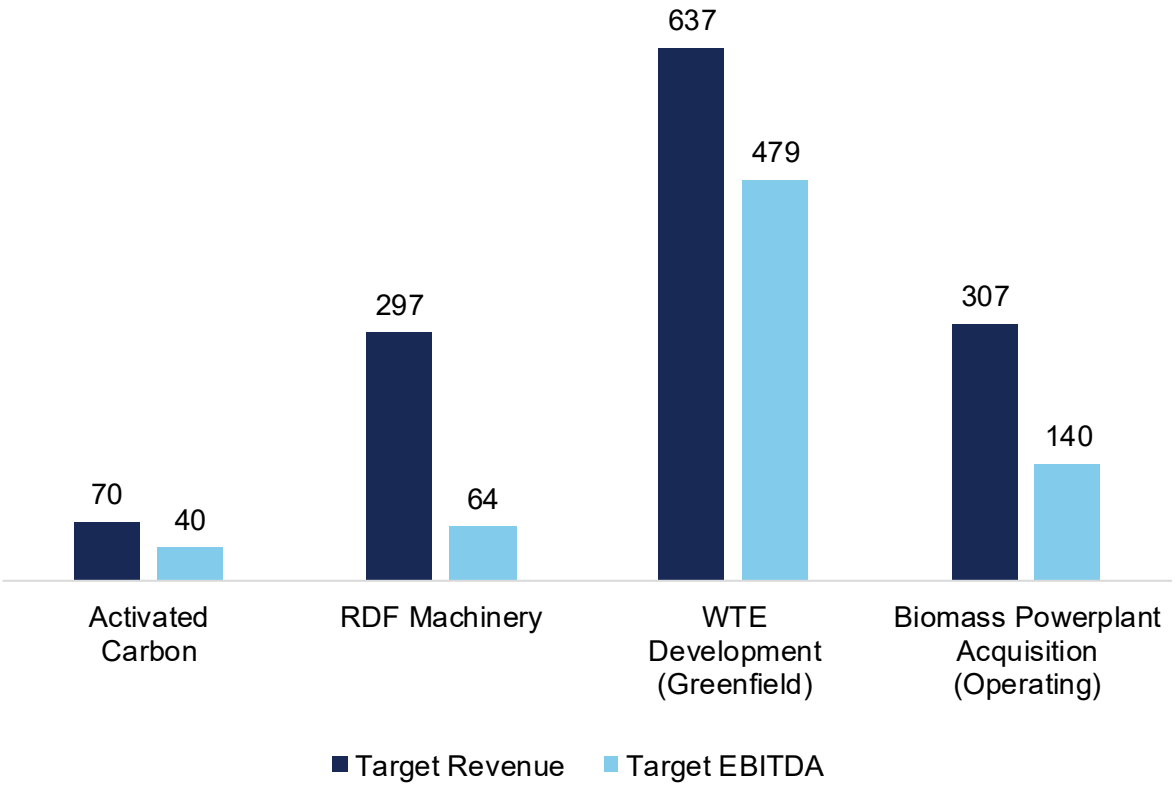
Target EBITDA



	Activated Carbon	Asset Light Driven	Delivers favorable performance compared to investment size
	RDF	Volume Driven	Drive the majority of revenue contribution
	WTE	Contract stability	long-term contracted and stable cash flow profile
	Biomass Acquisition	Cash Flow Today	Already recognized operating cash flow

Target Revenue and EBITDA by Project

Clear EBITDA uplift from higher-margin businesses

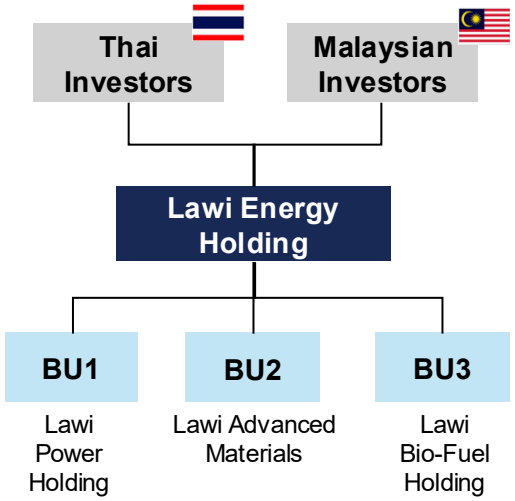


Investment Structure Option

Flexible investment structures across platform and project levels to match investor risk-return preferences

Minority Investment

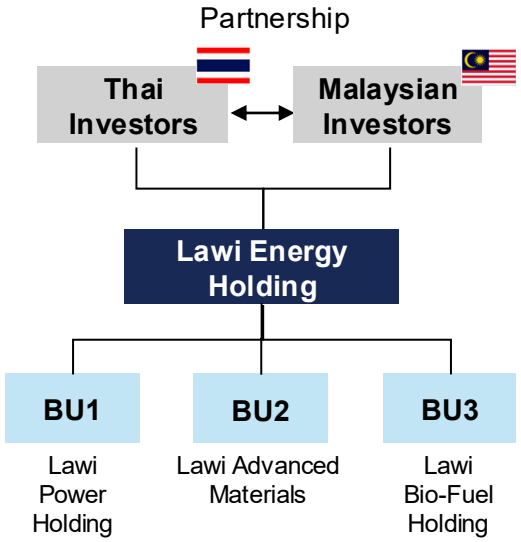
Passive exposure to diversify platform with limited downside



Minority stake at platform level providing exposure to a diversified portfolio of project investment.

Joint Venture (JV)

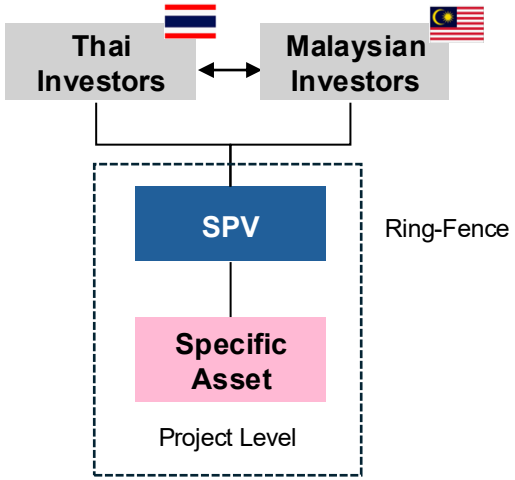
Balanced risk-return with shared governance and strategic alignment



Co-investment with Lawi at platform or project level, enabling shared governance and aligned execution with strategic partners

Project Level (SPV)

Asset-level investment with visible cash flows and ring-fenced risk



Direct investment into ring-fenced project SPVs with contracted revenue streams and asset-level cash flow visibility

Remark: for UPT Acquisition Project, we prefer investment structure as a Project Level but subject to further negotiated

Risk Mitigation Framework

Multi-layered downside protection through contracts, proven technology, and platform diversification

Key Risk Categories	Mitigation Strategies
Feedstock Supply Limited or volatile biomass supply across regions	**Diversified Supply Contracts** Multi-source sourcing strategy with long-term supply agreements across key regions
Revenue Risk Dependence on offtake contracts and demand visibility	**Secured Offtake Agreements** Long-term PPA and industrial offtake agreements with anchor customers
Technology Risk Processing efficiency and equipment reliability	**Proven Technology Platform** Use of established, operating technologies with Lawi Engineering Malaysia, Global Green Holding and KK Green
Execution & Development Risk Construction delays and cost overruns	**Phased Rollout Strategy** Modular expansion with experienced EPC partners and staged capital deployment
Price / Market Risk Exposure to commodity price fluctuations impacting margins	**Diversified Revenue Mix** Balanced exposure across contracted power plants, RDF Plants, and activated carbon products

De-risked through contractual visibility, proven operations, and diversified platform structure

Integrated ESG Impact

Delivering measurable ESG impact across waste, energy, and carbon reduction

End-to-End Impact Platform

Waste Diversion

(Environmental)

Diverts significant waste from landfills, cutting methane emissions at source

Coal Displacement

(Decarbonization)

RDF replaces coal, reducing carbon intensity for industrial users.

Import Substitution

(Industrial Resilience)

Local activated carbon production reduces reliance on imports.

Carbon Reduction

(Climate Impact)

Delivers measurable CO_2 reduction aligned with regional net-zero targets.

Social Value

(Social Impact)

Creates local jobs and supports green skill development

"Combining waste management, clean energy, and industrial decarbonization into a single integrated platform"

THANK YOU